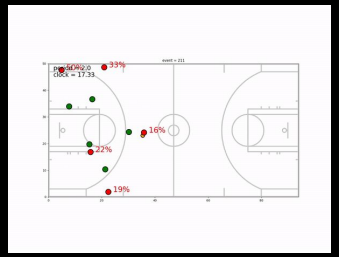
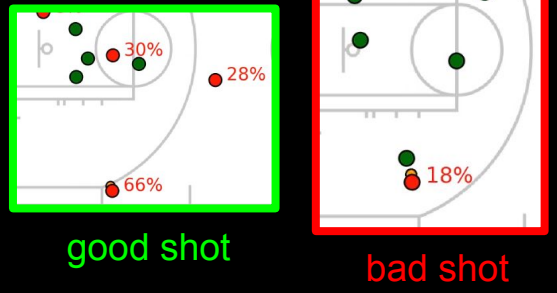
Focusing on the XY data for sports

Anson Wong

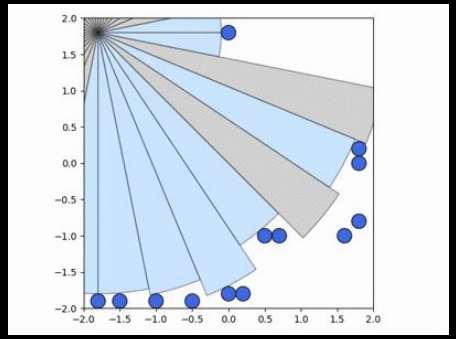
What I am doing these days

1) Shot-quality model (uses XY data)



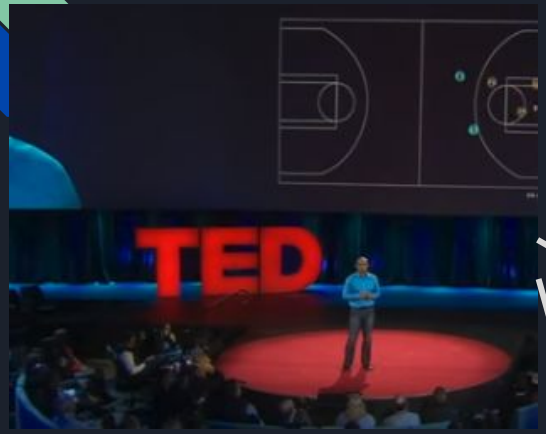


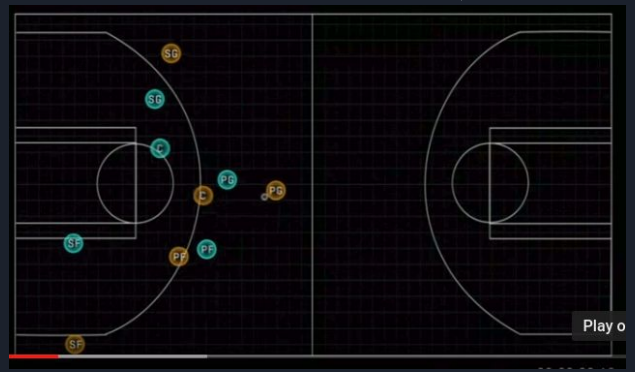
2) Agent-model to generate XY game strategies



Learn to develop strategies with a computer automatically!

TED talk (July 2015) about state-of-the-art NBA analytics

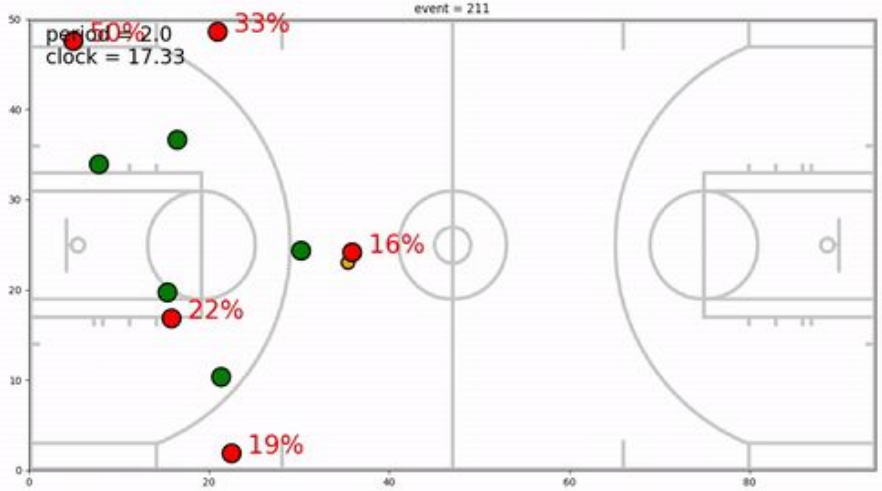




Obtain a measure of ‘shot quality’



What I built: a shot-quality automator

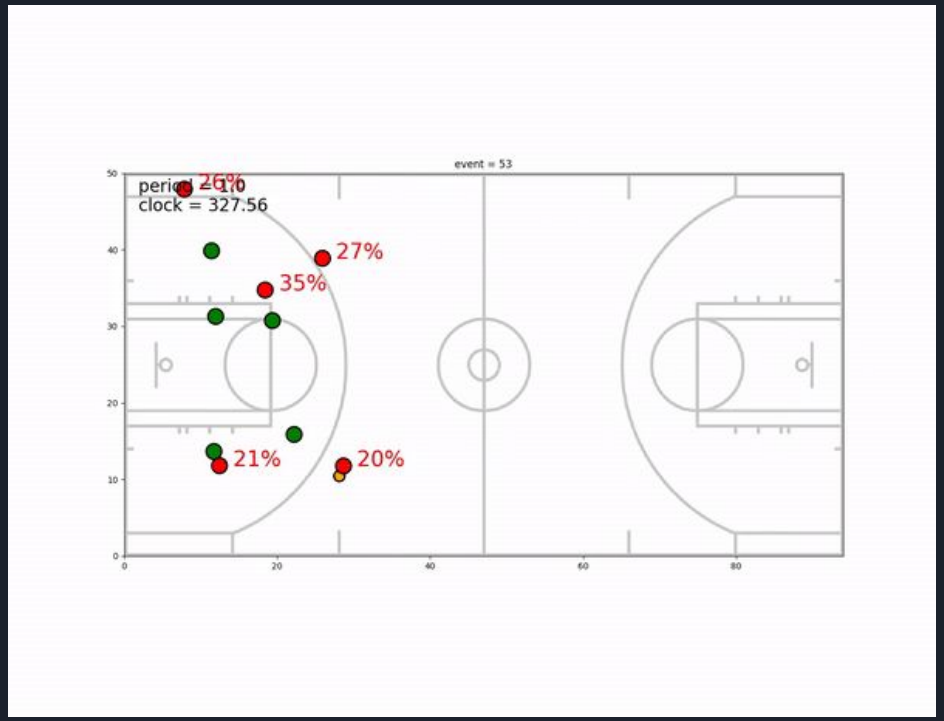


The XY data of an Atlanta vs Detroit NBA game

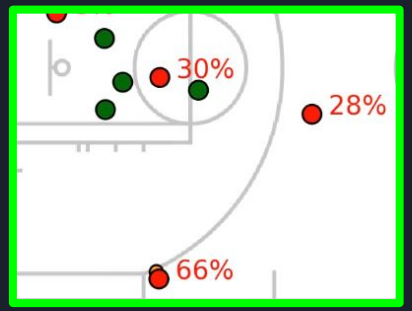
The % represent shot-quality here

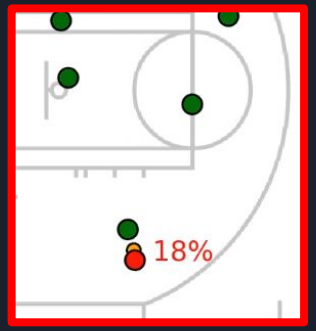
What I built: a shot-quality automator

Using XY + play-by-play data -> our player shot quality

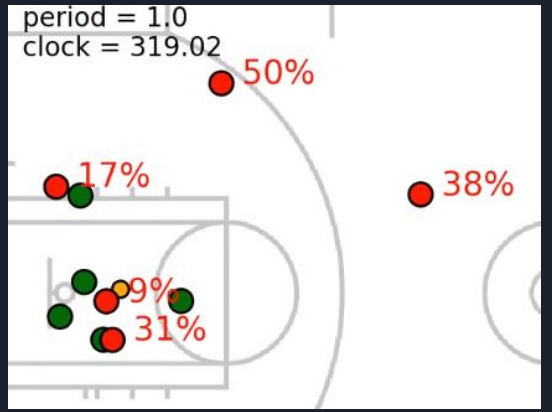


Shot % based on *defense* + *shot distance*





Were there better options???...



More about our shot-quality model:

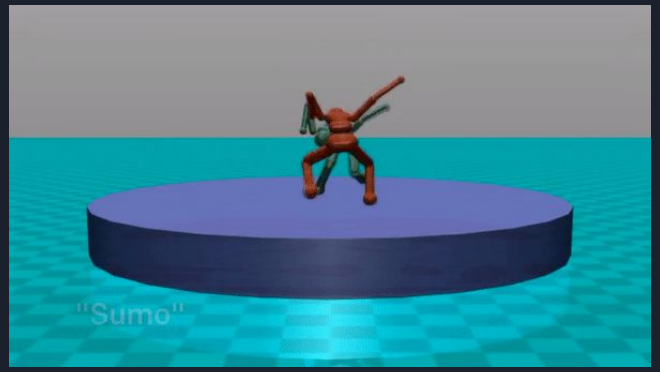
- Can be upgrade to be more player-specific (i.e. good/bad shooters and defences)
- Define a “defensive” metric for players or formations
- Automatically spot missed opportunities to help correct players
- Real-time shot analysis: we can say quickly tell you which plays were historically advantageous and disadvantageous

But can we do more with XY models?

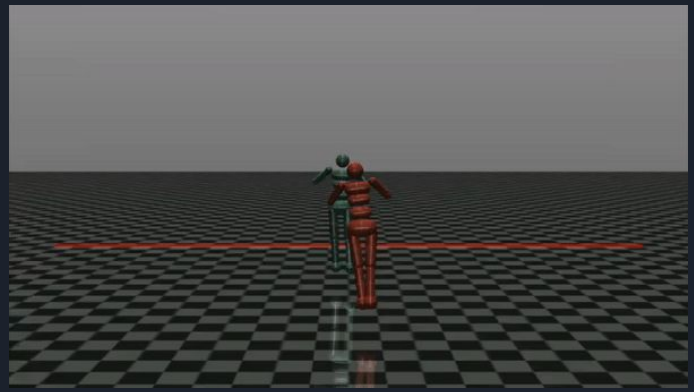
Let’s look at what’s out there:

OpenAI’s “Emergent Complexity via Multi-agent cooperation” paper

Self-play -> Sumo/soccer



Run past someone



But can we do more?

So the idea I had was to combine:

i) Self-play and deep reinforcement learning
ii) Common rewards/rules of sports games
iii) Actual XY data from sports matches

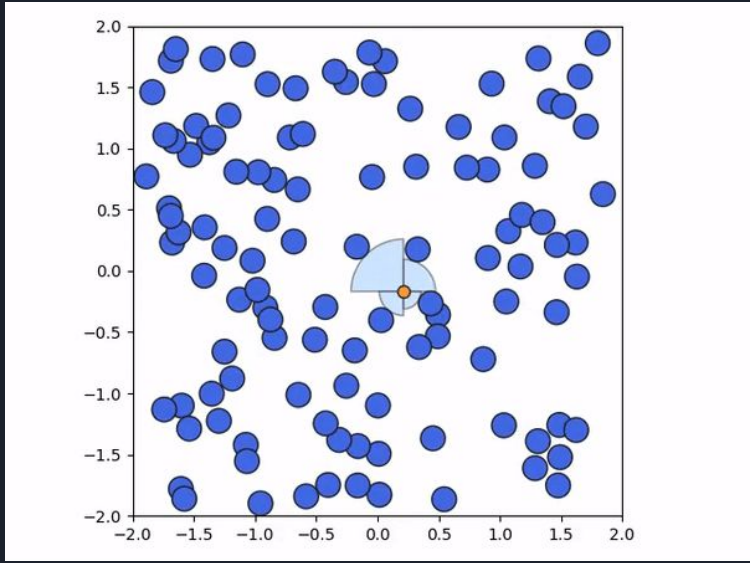
----> build a model that generates “optimal” strategies to win your game, and a way to evaluate player decisions (did they make the “right” play?)

------> XY Universe!!!

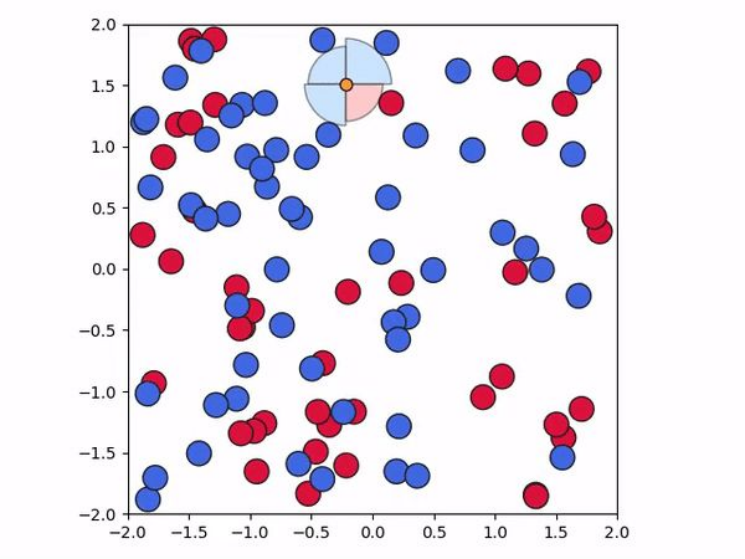
XY Universe: 2D particle strategies

Start by building the basics:

Avoid bad particles



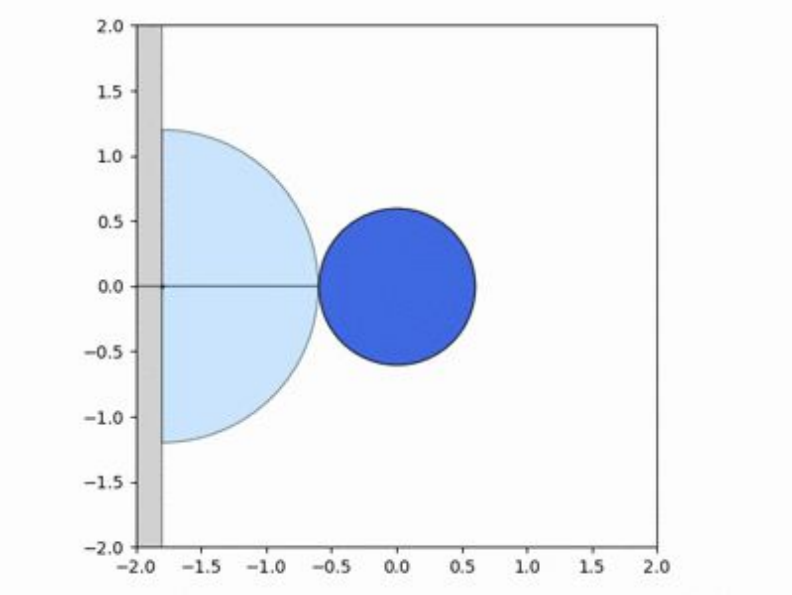
Avoid bad particles
and eat good particles



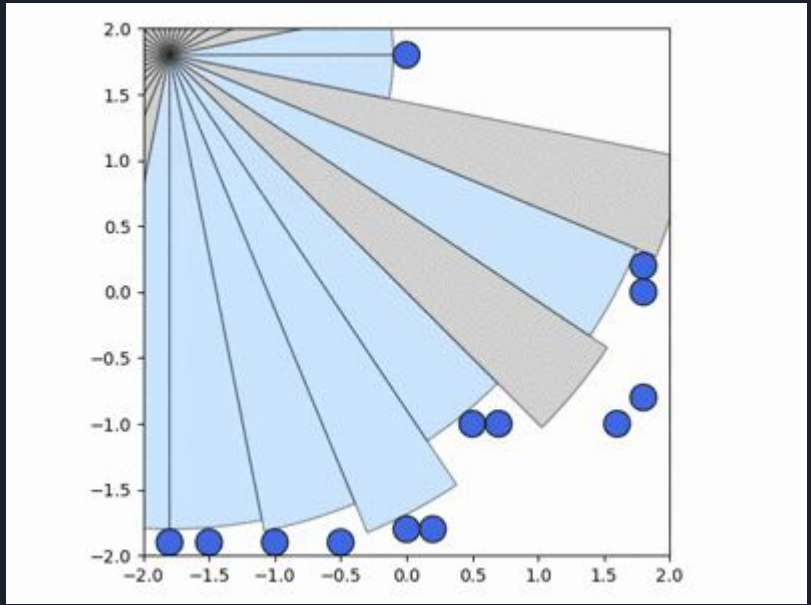
XY Universe: 2D particle strategies

Then a little more-specific...

Avoid bad particle and get to the right asap



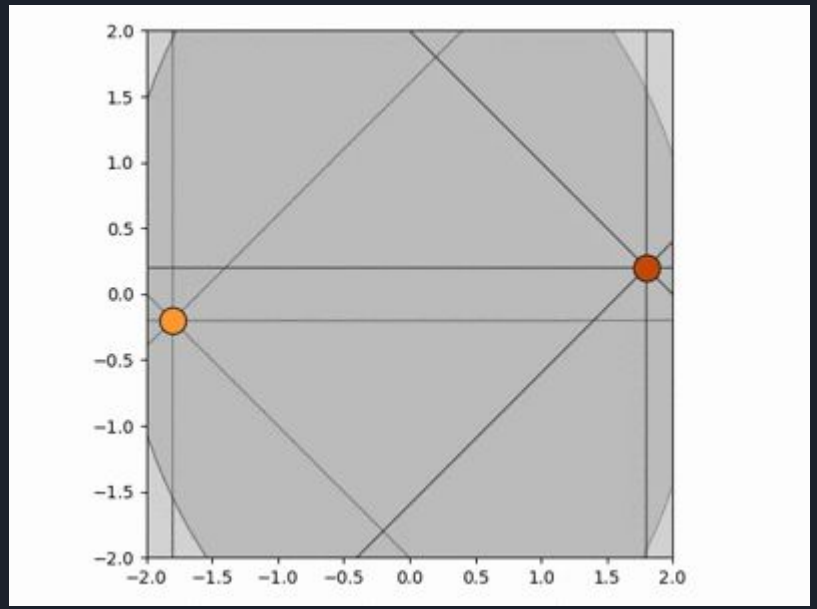
Avoid bad particles and get to the right asap



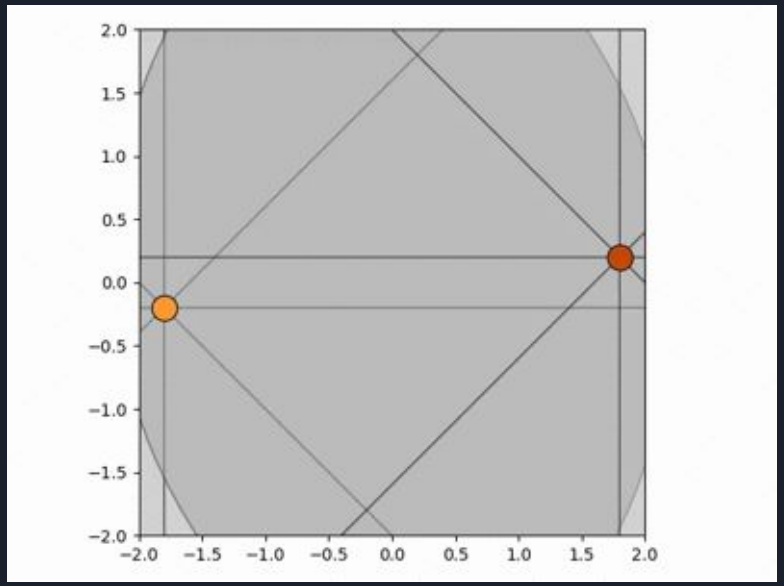
XY Universe: 2D particle strategies

Now add in some competitive self-play...

Get to your goal (early stages)

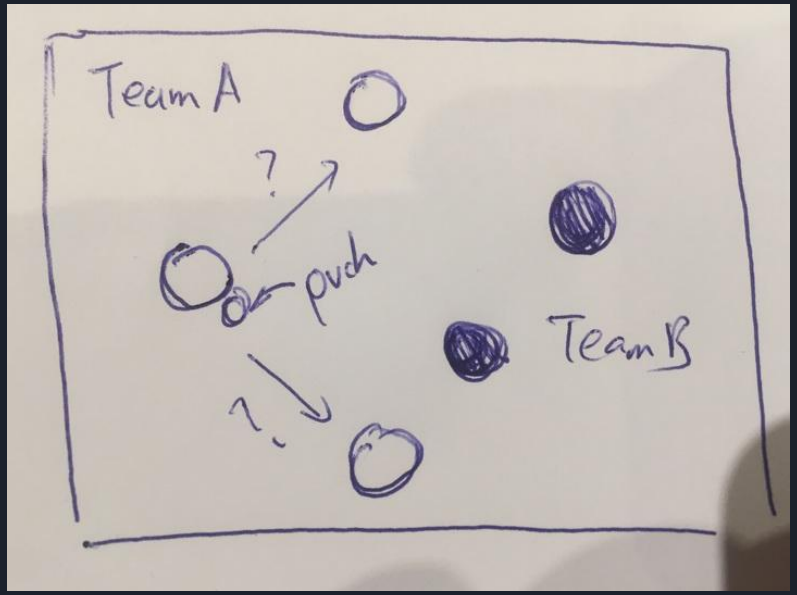


Get to your goal (late stages)



XY Universe: 2D particle strategies

What's in the future?…
-> add more complexity, more agents + passing



future

What this all means:

- Find strategies and formations based on weaknesses of opponents
- Study tendencies
- Correct bad habits of players
- Evaluate player skill for trades/draft